HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** scripts to automate data extraction and transformation for manufacturing process analysis, reducing manual effort by **~40%**.
- Assisted in the integration of **PLC** and **SCADA** data streams with a central **MES** historian for a pharmaceutical client's production line, enabling real-time operational visibility.
- Contributed to the development of **AI-driven applications** focused on predictive maintenance by structuring and analysing historical operational data, aiming to reduce unscheduled downtime.
- Supported the implementation of **digital twin** concepts by collecting and processing sensor data from manufacturing equipment for simulation and analysis.
- Participated in the testing and validation of **automation systems** within a GMP-regulated environment, ensuring compliance with stringent industry standards.
- Collaborated with **OEMs** and internal **data scientists** to integrate new automation hardware and AI platforms into existing manufacturing workflows.
- Managed **change control** processes for system configuration updates and supported post-implementation triage for digitalisation initiatives across interconnected enterprise systems.

SKILLS

Programming Languages – Python, SQL, Java, R

Cloud & Platforms – AWS (EC2, S3, Lambda, CloudWatch), Azure AI, Azure Machine Learning, Databricks

Automation & Control – PLC, SCADA, MES, Rockwell Automation (desirable)

AI & Machine Learning – Machine Learning, Data Analytics, Digital Twin, Predictive Maintenance, AI

Databases & Data Engineering – PostgreSQL, MySQL, Oracle, Data Warehousing, ETL, Data Modelling

BI & Visualization – Power BI, Tableau, DAX, Power Query

DevOps & Tools – Docker, Jenkins, Git, Jira, Confluence, Linux

PROJECTS

Manufacturing Process Optimisation Platform (HCL Technologies)

- Developed **Python** scripts to automate the ingestion and transformation of sensor data from manufacturing equipment, creating structured datasets for analysis.
- Built **SQL** queries against PostgreSQL to extract and aggregate operational metrics from the client's production database, supporting reporting on efficiency gains.
- Created interactive **Power BI** dashboards to visualise key performance indicators derived from the processed data, enabling stakeholders to monitor production throughput and identify bottlenecks.

Predictive Maintenance Model Development (MSc Applied Project)

- Engineered feature pipelines in **Python** using pandas and NumPy to extract relevant signals from historical sensor readings for predictive maintenance.
- Trained a **scikit-learn** Random Forest classifier on a dataset of ~150,000 sensor events to predict equipment failures, achieving **~0.92 AUC** on held-out test data.
- Deployed the trained model as a REST API using **FastAPI**, enabling real-time inference for potential failure alerts, with p95 latency under **200ms**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)